

Grama-Suvidha Portal

Village Infrastructure Transparency App
Comprehensive Project Requirement Document

1. Project Overview

Project Title	Grama-Suvidha Portal – A Village Development Tracking & Transparency App
Domain	Governance / Infrastructure / Civic Engagement
Tech Stack	Android Studio Kotlin XML UI Room DB LiveData MVVM JSON (Mock API)

Objective

Grama-Suvidha Portal is an Android app designed to provide transparency in rural infrastructure projects. It allows citizens to view ongoing government works, track progress, and give feedback.

Core Idea: "Make village development visible, trackable, and accountable to every citizen."

2. Purpose and Goals

Purpose

- Build a real-world governance app
- Demonstrate Android development skills
- Create a socially impactful project

Goals

- Display all village projects clearly
- Show real-time progress visually
- Enable citizen feedback
- Provide offline access
- Support Kannada and English

3. Scope of the Application

In-Scope Functionalities

- Project List Dashboard — View all ongoing village development works with progress status
- Project Detail Screen — Display project description, completion percentage, and status updates
- Citizen Feedback System — Allow users to rate projects and report issues/comments
- Persistent local data storage using Room Database
- Kannada & English language support for accessibility
- Mock API integration using local JSON data
- Image loading and project progress visualization using Glide/Coil
- Export project summary report as PDF/CSV (optional advanced feature)

Out of Scope (for this internship)

- Real-time government database/API integration
- Cloud-based multi-device synchronization
- GPS-based live project location tracking
- Automated AI-based project prediction system (These can be considered as future extensions.)

4. Target Users and Personas

Primary Users

- Village Citizens: People who want to track local development projects and infrastructure progress
- Panchayat Members: Officials responsible for monitoring and updating project status
- Rural Workers & Contractors: Individuals involved in project execution and maintenance

User Persona Example

Field	Details
Name	Ramesh Gowda
Age	42
Occupation	Farmer

Name: Yashaswini B S
USN: 1KT22AI062

Location	Rural Karnataka, India
Goal	Track village road and water supply project progress easily
Pain Points	No proper updates about government works and project completion status
Expectation	A simple app showing clear project progress, status, and updates

5. Functional Requirements

5.1. Project List Dashboard

- View all village development projects in a scrollable dashboard
- Tap a project card to open detailed project information
- Display project progress using visual progress bars
- Show project status such as Ongoing, Completed, or Pending
- Optional: display project images and latest update date

5.2. Project Progress Dashboard

- Pie Chart showing percentage breakdown of projects by status (e.g., 60% Completed, 25% Ongoing, 15% Pending)
- Bar Chart showing project completion percentage over the selected time period
- Summary cards: Total Projects | Completed Projects | Ongoing Projects
- Charts update dynamically whenever project progress/status changes

5.3. Project Status & Update Management

- Admin updates project progress percentage and current work status
- Project status automatically changes based on progress updates
- Status Alert: Visual warning for delayed or pending projects
- Option to add new project updates, images, and completion reports

5.4. Project Update Log

- View total completed, ongoing, and pending projects for Today / This Week / This Month
- Filterable project history by date range or project status
- Each entry shows: Project Name | Status | Progress % | Update Date | Location
- Monthly summary card: Total Projects | Completed Projects | Delayed Projects

5.5. GenAI – Smart Project Advisor

- Gemini API integration: Analyzes project progress and generates natural-language suggestions
- Example output: “Road construction project is delayed by 2 weeks. Consider increasing workforce allocation to complete work on time.”
- Appears as an Advisor card on the dashboard — refreshes whenever project status or progress is updated

5.6. Project & Category Management

- Add new village development projects with project name, category, and location details
- Set project status, estimated completion date, and progress percentage
- Archive completed or cancelled projects without deleting previous project history

6. Non-Functional Requirements

Requirement	Description
Performance	Dashboard and project details must load within 2 seconds.
Usability	Simple UI with large cards and minimal user interaction for easy accessibility.
Reliability	No data loss on app restart; Room DB ensures persistent local storage.
Portability	Compatible with Android 9.0 (API 28) and above.
Scalability	Architecture must support future Firebase integration and real-time government APIs.
Security	Local project and feedback data stored securely within the device.
Offline Support	Core functionalities work without internet; GenAI feature degrades gracefully offline.

7. System Architecture

Architecture Pattern: MVVM (Model-View-ViewModel)

The app follows the MVVM architectural pattern to ensure separation of concerns, maintainability, and clean scalability for future enhancements.

Layer	Technology	Role
View (UI)	XML Layouts / Jetpack Compose	Renders screens and displays project information
ViewModel	Kotlin ViewModel + LiveData	Handles UI state and business logic
Repository	Repository Pattern	Single source of truth; manages local data and APIs
Local DB	Room Database	Persistent CRUD operations for projects and feedback
GenAI Layer	Gemini API (REST)	Analyzes project progress and generates smart recommendations
Charts	MPAndroidChart	Pie/Bar chart visualization for project analytics dashboard

Data Flow

- User (Project Update / Feedback Entry) → UI (View) → ViewModel → Repository → Room Database
- Dashboard charts pull aggregated project data from Repository → ViewModel → UI
- GenAI Advisor reads project summaries from Repository and calls Gemini API → result displayed on dashboard

8. Database Schema

Table	Fields	Description
Project	id (PK), name, description, status, progress_percentage, location	Stores details of village development projects
Feedback	id (PK), project_id (FK), rating, comment, date	Stores citizen feedback and project reviews
ProjectUpdate	id (PK), project_id (FK), update_title, update_description, update_date	Stores project progress updates and activity logs
Analytics	id (PK), total_projects, completed_projects, pending_projects	Stores aggregated dashboard analytics data
AIAdvice	id (PK), generated_at, advice_text, project_snapshot	Cached GenAI-generated project recommendations

9. User Flow Diagram (Text Outline)

- 1. Splash Screen → Home Dashboard
- 2. Home Dashboard → Project Progress Chart + Delayed Project Alerts + Project Summary + AI Advice Card

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- 3. Project List Screen → Select Project Card → View Project Details → Track Progress → Submit Feedback
- 4. Analytics Screen → Pie Chart (by project status) + Bar Chart (by completion percentage) + Filter by date range
- 5. Project Update Log → This Week / This Month toggle → Project activity history list
- 6. Project Manager Screen → View all projects + status updates → Edit progress / update details
- 7. Settings → Language preference (Kannada/English) → Export PDF/CSV → About App

10. UI/UX Guidelines

Design Goals

- Accessibility-first: Large cards, buttons, and simple navigation for easy usability
- Visual decision-making: Progress bars, colors, and charts replace text-heavy information
- Minimal text entry — quick project viewing and feedback submission are the core interactions
- Clean governance-inspired aesthetic: Blue, green, and white theme representing transparency and development.

Color Palette

Role	Color	Hex Code	Usage
Primary	Royal Blue	#1976D2	App bar, buttons, headings
Accent	Green	#4CAF50	Highlights, progress indicators, action buttons
Success	Light Green	#66BB6A	Completed project indicators
Warning	Red	#E53935	Delayed or pending project alerts
Background	Soft White	#F5F5F5	Screen backgrounds

Key Screens

- 1. Splash / Intro Screen – App logo + tagline + loading animation
- 2. Home Dashboard – AI Advice Card + Project Progress mini-chart + Delayed Project Alerts + Quick Access Buttons
- 3. Project List Screen – Project card list + status indicator + progress bar + View Details button
- 4. Analytics Screen – Pie chart (project status breakdown) + Bar chart (completion percentage) + date filter

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- 5. Project Update Log Screen – Week/Month toggle + scrollable project update history + summary card
- 6. Project Manager Screen – Project cards showing status + progress update button + alert indicators
- 7. Settings Screen – Language preference + export PDF/CSV + About App

11. Analytics and Reports

- Use MPAndroidChart library for all chart rendering
- Pie Chart: Project status distribution (e.g., Completed 60%, Ongoing 25%, Pending 15%)
- Bar Chart: Project completion percentage across the selected time period
- Line Chart (optional): Project progress trend over the past 4 weeks
- Filters: Today | This Week | This Month | Custom Range
- Top 3 completed projects highlighted with trophy/star badge
- GenAI Insight: Natural-language summary of project trends and delays using Gemini API

12. Reference Apps

App	Description	Reference
mSeva Karnataka	Government citizen service platform	Karnataka Government Portal
MyGov India	Citizen engagement and governance platform	MyGov India
Swachhata-MoHUA	Public issue reporting and civic management app	Swachhata Portal
Karnataka One	Unified citizen services and public utility platform	Karnataka One

13. Deliverables for Students

- 1. Wireframe (Figma/Canva) – All 7 key screens with complete navigation flow
- 2. Database Schema Diagram – ER model showing all project-related tables and relationships
- 3. Functional App (APK) – Working prototype with Project Dashboard + Progress Tracking + Feedback System

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- 4. Analytics Dashboard – Pie/Bar charts updating dynamically based on project data
- 5. GenAI Integration – Gemini API advisor card displaying smart project recommendations
- 6. Project Report – Detailed documentation with features, screenshots, architecture, and code overview
- 7. Demo Video (2–3 min) – Screen recording explaining all major app functionalities

14. Impact Goals

Transparency & Accountability

Providing citizens with clear visibility into village development projects, progress status, and infrastructure updates.

Public Engagement

Encouraging villagers to actively monitor government works and submit feedback through a simple digital platform.

Efficient Project Monitoring

Helping local authorities track project completion, delays, and pending works more effectively.

GenAI Empowerment

Making AI-powered project insights accessible through simple natural-language recommendations and status summaries.

15. Future Scope & Enhancement Ideas

- Firebase Cloud Sync – Multi-device access and automatic project data backup
- GPS-Based Project Tracking – Live location tracking for village infrastructure works
- WhatsApp Share – Share weekly project progress reports directly via WhatsApp
- Multi-Admin Mode – Multiple Panchayat members managing project updates collaboratively
- Regional Language Support – Kannada, Hindi, Tamil, and Telugu language support
- Push Notifications – Alerts for delayed projects and new updates
- Advanced AI Model – Predict project delays and recommend resource allocation using historical data

16. Evaluation Parameters

Category	Weightage
Functionality (Project Dashboard, Progress Tracking, Feedback System, Analytics)	30%
UI/UX Design (Accessibility, Visual Appeal, Ease of Use)	25%
Code Quality & Structure (MVVM, clean code, Room DB)	20%
Innovation / GenAI Integration (Gemini API Advisor)	15%
Presentation & Documentation (Report + Demo Video)	10%

17. Success Criteria

- The Project Progress chart must update instantly whenever project status changes
- The Project Update Log must allow filtering by “This Week” and “This Month”
- Delayed Project Alerts must be displayed prominently on the dashboard
- Project details and feedback flow must be completable with minimal user interaction
- The GenAI Advisor card must display meaningful recommendations based on actual project data
- The app must function fully offline (except GenAI feature)

18. Next Action Items for Teams

- Define complete project catalogue — project names, categories, locations, and default status values
- Prepare wireframes for all 7 screens in Figma and get mentor approval
- Divide responsibilities: UI Developer | Database Handler | GenAI Integrator | Tester
- Create Android project in Android Studio with MVVM folder structure
- Obtain Gemini API key from Google AI Studio for GenAI integration

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